

Yang Zhou

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[Personal Website](#) | [LinkedIn](#) | [Google Scholar](#) | [GitHub](#) | [Hugging Face](#)

EDUCATION

Rutgers University September 2023 – Present
Ph.D. in Computer Science
Specialization: Large Language Models, Computer Vision

University of Science and Technology of China September 2020 – June 2023
Master of Engineering, Control Science and Engineering
Specialization: Computer Vision, Object Detection, CNN Architecture

Southwest University September 2016 – June 2020
Bachelor of Engineering, Automation
Specialization: Signals & Systems

EXPERIENCE

TikTok Inc. May 2026 – Present
Research Scientist Intern, San Jose, CA

- Built a self-evolving agent framework that externalizes experience into a retrievable asset pool of memories, skills and tools, and maintains it at scale through hybrid recall, write-time deduplication, scope revision and consolidation into skills.
- Formalized evolution as a coverage-versus-compression trade-off, with a stopping criterion on its marginal return, the boundary at which further evolution stops paying, and the optimal storage granularity as a function of retriever quality.
- Characterized the decision problem in LLM memory maintenance: unconditional erasure makes collapse structural, so repairs narrow an asset’s applicability boundary against a frozen base rather than rewrite or delete it.

PUBLICATIONS

MLLM & LLM

- **Evaluating LLMs When They Do Not Know the Answer: Statistical Evaluation of Mathematical Reasoning via Comparative Signals**
Zihan Dong, Zhixian Zhang, **Yang Zhou**, Can Jin, Ruijia Wu, Linjun Zhang.
[ICML 2026](#).
- **LED: LLM Enhanced Open-Vocabulary Object Detection without Human Curated Data Generation**
Yang Zhou, Shiyu Zhao, Yuxiao Chen, Zhengting Wang, Can Jin, Dimitris N. Metaxas.
[CVPR 2026](#). [Website](#).

Multi-Agent

- **Evidence Over Plans: Online Trajectory Verification for Skill Distillation**
Yang Zhou, Zihan Dong, Zhenting Wang, Can Jin, Shiyu Zhao, Bangwei Guo, Difei Gu, Linjun Zhang, Mu Zhou, Dimitris N. Metaxas.
[arXiv, 2026](#). [NeurIPS 2026 Submit](#). [Website](#).
- **M³-Bench: Multi-Modal, Multi-Hop, Multi-Threaded Tool-Using MLLM Agent Benchmark**
Yang Zhou, Mingyu Zhao, Zhenting Wang, Difei Gu, Bangwei Guo, Can Jin, Ligong Han, Shiyu Zhao, Mu Zhou, Dimitris N. Metaxas.
[arXiv, 2025](#). [NeurIPS 2026 Submit](#). [Website](#).

- **CAMEL: A Framework for Finding the Scaling Laws of Agents**

Ziyi Yang, ... **Yang Zhou**, ...
[Website](#), [GitHub](#).

RL

- **DARE: Difficulty-Adaptive Reinforcement Learning with Co-Evolved Difficulty Estimation**

Yang Zhou, Can Jin, Zihan Dong, Zhepeng Wang, Yanting Yang, Shiyu Zhao, Lei Li, Runxue Bao, Yaochen Xie, Dimitris N. Metaxas.

[arXiv, 2026](#) . [NeurIPS 2026 Submit](#) . [Website](#).

- **Your Reward Function for RL is Your Best PRM for Search: Unifying RL and Search-Based TTS**

Can Jin, **Yang Zhou**, Qixin Zhang, Hongwu Peng, Di Zhang, Marco Pavone, Ligong Han, Zhang-Wei Hong, Tong Che, Dimitris N. Metaxas.

[arXiv, 2025](#) .

Computer Vision & Medical Imaging

- **RadAlign: Advancing Radiology Report Generation with Vision-Language Concept Alignment**

Difei Gu, Yuxuan Gao, **Yang Zhou**, Mengmi Zhou, Dimitris Metaxas.

[MICCAI 2025](#) .

- **A Multimodal Spatio-Temporal GCN Model with Enhancements for Isolated Sign Recognition**

Yang Zhou, Zhaoyang Xia, Yuxiao Chen, Carol Neidle, Dimitris N. Metaxas.

[LREC-COLING 2024](#) .

- **A Review of Convolutional Neural Network Architectures and Their Optimizations**

Shuang Cong, **Yang Zhou**.

[Artificial Intelligence Review](#) [IF 9.588/Q1](#) .

- **Diffusion Models for Sign Language Video Anonymization**

Zhaoyang Xia, **Yang Zhou**, Ligong Han, Carol Neidle, Dimitris N. Metaxas.

[LREC-COLING 2024](#) .

Other Collaborative Work

- **UniPro: Unified Multi-Mode Medical Image Segmentation from 2D Images to 3D Volumes via Propagation**

Bangwei Guo, Yunhe Gao, Meng Ye, **Yang Zhou**, Difei Gu, Guoning Zhang, Leon Axel, Dimitris N. Metaxas.

[NeurIPS 2026 Submit](#) .

- **LUCID-SAE: Learning Unified Vision-Language Sparse Codes for Interpretable Concept Discovery**

Difei Gu, Yunhe Gao, Gerasimos Chatzoudis, Zihan Dong, Guoning Zhang, Bangwei Guo, **Yang Zhou**, Mu Zhou, Dimitris N. Metaxas.

[arXiv, 2026](#) .

- **K-Prism: A Knowledge-Guided and Prompt Integrated Universal Medical Image Segmentation Model**

Bangwei Guo, Yuxuan Gao, Mengqi Ye, Difei Gu, **Yang Zhou**, Leon Axel, Dimitris Metaxas.

[arXiv, 2025](#) .

Previous Work (Modeling, Data Analysis, Signals & Systems)

- **A Multistory Building Evacuation Model Based on Multiple-Factor Analysis**

Yang Zhou, Zichuan Fan.

[Advances in Civil Engineering](#) [IF 1.924/Q3](#) .

- **The Excitation and Detection of Lamb Waves in a Droplet-Loaded Plate Using Air-Coupled Ultrasonic Transducers**

Zichuan Fan, **Yang Zhou**, Tanghong Wu.

Measurement IF 5.131/Q1 .

- **Quasi-Dispersion of Air-Coupled Ultrasonic Signal for Angle-Dependent Reception**

Zichuan Fan, **Yang Zhou**, Tianhao Qie.

Measurement IF 5.131/Q1 .

- **Multiphysics Model of Lamb Waves Propagation in a Plate Loaded with Droplets**

Yang Zhou, Zichuan Fan.

ICCAR 2019 IEEE .

- **Multiple Reflective Signal Reception in Gas Flow Measurement Using Air-Coupled Leaky Lamb Waves**

Zichuan Fan, Tianhao Qie, **Yang Zhou**.

Measurement IF 5.131/Q1 .

TEACHING & TALKS

Computer Security (Teaching Assistant)

Spring 2025

Rutgers University, New Jersey, USA

Systems Programming (Teaching Assistant)

Fall 2023, Spring 2024

Rutgers University, New Jersey, USA

Neural Networks and Deep Learning (Teaching Assistant)

Aug 2021

University of Science and Technology of China, Hefei, China

Characteristics of Lamb Waves and Their Leakage Waves (Presenter)

May 2019

ICCAR International Academic Conference, Beijing, China

SKILLS

- **Operating Systems:** Windows, Linux
- **Programming:** Python (primary), C++, Java
- **Tools:** Office Suite, MATLAB

HONORS & PATENTS

- 2019 Invention Patents (China)
- 2019 Utility Model Patent (China)
- 2019 Mathematical Contest in Modeling (COMAP) – Meritorious Winner